# IP Addressing & Subnetting Assignment

Comprehensive Long-Form Technical Report & In-Depth Solutions

## Task 1: IP Address Classes (A, B, C, D, E) & Real-World Application Mapping

An IPv4 address is a 32-bit numerical label divided into four 8-bit octets. Historically, under the Classful Addressing architecture (RFC 791), IPv4 addresses were categorized into five distinct classes (A through E) based on the leading bits of the first octet.

| Class | 1st Octet Range | Default Subnet Mask | Network / Host Split | Real-World Example Device or Service |
|---|---|---|---|---|
| **Class A** | 1 – 126 | 255.0.0.0 (/8) | 8 Bits Net / 24 Bits Host | Large global ISPs, multinational telecom backbones (e.g., AT&T, IBM corporate network). |
| **Class B** | 128 – 191 | 255.255.0.0 (/16) | 16 Bits Net / 16 Bits Host | Large university campus networks (e.g., University WiFi infrastructure), mid-sized enterprises. |
| **Class C** | 192 – 223 | 255.255.255.0 (/24) | 24 Bits Net / 8 Bits Host | Small office / Home routers (SOHO LANs, e.g., home Wi-Fi connecting phones & laptops). |
| **Class D** | 224 – 239 | N/A (Multicast) | Multicast Group Addressing | IPTV video broadcasting streams, routing protocol updates (e.g., OSPF multicast 224.0.0.5). |
| **Class E** | 240 – 255 | N/A (Reserved) | Experimental / Research | IETF experimental setups, future research testing, and specialized networking benchmarks. |

*Note: 127.0.0.0/8 is reserved for internal host loopback testing (e.g., 127.0.0.1 for localhost).*

## Task 2: Detailed Network Calculation for IP 192.168.10.45 /24

**Given Inputs:**
• Target IP Address: 192.168.10.45
• Subnet Mask: 255.255.255.0 (CIDR Prefix: /24)

### Step-by-Step Calculation Steps:

- **Step 1: Binary Representation & Subnet Mask Analysis**
  The subnet mask 255.255.255.0 in binary consists of 24 consecutive '1' bits followed by 8 '0' bits:

```
IP Address (192.168.10.45): 11000000 . 10101000 . 00001010 . 00101101
Subnet Mask (255.255.255.0): 11111111 . 11111111 . 11111111 . 00000000
```

- **Step 2: Calculate Network Address (Bitwise AND Operation)**
  Performing a logical bitwise AND between the IP address and the subnet mask sets all host bits to 0:

```
11000000.10101000.00001010.00101101 (192.168.10.45)
AND 11111111.11111111.11111111.00000000 (255.255.255.0)
-------------------------------------------------
= 11000000.10101000.00001010.00000000 = 192.168.10.0
```

➔ **Network Address:** 192.168.10.0

- **Step 3: Calculate Broadcast Address**
  Setting all 8 host bits in the last octet to binary '1' (or binary OR with inverse mask):

```
11000000.10101000.00001010.11111111 = 192.168.10.255
```

➔ **Broadcast Address:** 192.168.10.255

- **Step 4: Determine Range of Valid Usable Host Addresses**
  The valid host range spans from one address above the network address to one address below the broadcast address:

```
First Usable Host = 192.168.10.0 + 1 = 192.168.10.1
Last Usable Host = 192.168.10.255 - 1 = 192.168.10.254
```

➔ **Usable Host Range:** 192.168.10.1 to 192.168.10.254 (Total 254 usable host addresses).

> **Summary of Final Results for Q2:**
> • Network Address: 192.168.10.0
> • Broadcast Address: 192.168.10.255
> • Range of Valid Host Addresses: 192.168.10.1 – 192.168.10.254

## Task 3: Subnetting Plan for Food Delivery Startup (Zomato)

**Requirement:** Divide the base network 10.0.0.0/24 into at least **4 equal, separate subnets** for Delivery Partners, Restaurants, Customers, and Admins.

### Subnet Calculation Methodology:

- **Base Network:** 10.0.0.0/24 (Total $2^8 = 256$ IP addresses).
- **Subnets Needed:** 4 subnets $ ightarrow$ Number of borrowed subnet bits $n = \log_2(4) = 2$ bits.
- **New CIDR Prefix:** Original prefix $/24 + 2 ext{ borrowed bits} = \mathbf{/26}$.
- **New Subnet Mask:** In binary, 26 ones equals 11111111.11111111.11111111.11000000 $ ightarrow$ 255.255.255.192.
- **Block Size (Increment):** $256 - 192 = 64$ IPs per subnet (62 usable host IPs per subnet).

### Comprehensive Subnet Allocation Plan Table:

| Group Name | Subnet Network Address | Subnet Mask (CIDR) | Usable Host IP Range | Broadcast Address |
|---|---|---|---|---|
| **1. Delivery Partners** | 10.0.0.0 | 255.255.255.192 (/26) | 10.0.0.1 – 10.0.0.62 | 10.0.0.63 |
| **2. Restaurants** | 10.0.0.64 | 255.255.255.192 (/26) | 10.0.0.65 – 10.0.0.126 | 10.0.0.127 |
| **3. Customers** | 10.0.0.128 | | | 10.0.0.191 |

| Group Name | Subnet Network Address | Subnet Mask (CIDR) | Usable Host IP Range | Broadcast Address |
|---|---|---|---|---|
| | | 255.255.255.192 (/26) | 10.0.0.129 – 10.0.0.190 | |
| **4. Admins** | `10.0.0.192` | 255.255.255.192 (/26) | 10.0.0.193 – 10.0.0.254 | 10.0.0.255 |

## Task 4: College Wi-Fi Subnetting Scenario (172.16.0.0/16)

**Scenario Constraints:**
• Base Network: `172.16.0.0/16`
• Requirement: Create at least **20 subnets** for campus buildings.

### Step-by-Step Subnetting Calculation:

- **Step 1: Calculate Borrowed Bits Needed for 20 Subnets**
  Using the formula $2^n \ge ext{Subnets Required}$, where $n$ is the number of borrowed bits:
  • For $n = 4$: $2^4 = 16$ subnets (Insufficient, less than 20).
  • For $n = 5$: $2^5 = 32$ subnets (Sufficient, accommodates 20 buildings with 12 spare subnets for future growth).
  ➔ **Borrowed Bits ($n$):** 5 bits.
- **Step 2: Calculate New Subnet Mask & Prefix**
  • New CIDR Prefix = Original $/16 + 5 ext{ bits} = \mathbf{/21}$.
  • Binary Mask (21 ones): `11111111 . 11111111 . 11111000 . 00000000`
  • Converting 3rd octet to decimal: $128 + 64 + 32 + 16 + 8 = 248$.
  ➔ **New Subnet Mask:** `255.255.248.0`
- **Step 3: Calculate Available Hosts Per Subnet**
  • Remaining Host Bits $h = 32 - 21 = 11$ bits.
  • Total Addresses per Subnet = $2^{11} = 2,048$ IPs.
  • Formula for Usable Hosts = $2^h - 2$ (subtracting Network and Broadcast addresses).
  • Usable Hosts = $2,048 - 2 = 2,046$ hosts.
  ➔ **Hosts Available per Subnet:** `2,046 usable host IPs`.

> **Final Answers for Task 4:**
> • **New CIDR Prefix:** `/21`
> • **New Subnet Mask:** `255.255.248.0`
> • **Total Subnets Created:** 32 subnets (exceeds the 20 minimum requirement)
> • **Usable Hosts Per Subnet:** `2,046 hosts`

## Task 5: Troubleshooting Invalid IP Address Scenario (192.168.1.300)

**Friend's Issue:** Device configured with IP address `192.168.1.300` cannot connect to the network.

> **1. Identification of Error:**
> The IP address `192.168.1.300` is **invalid and syntactically illegal**. In IPv4 addressing, each of the four octets is an 8-bit unsigned integer, meaning its decimal value must fall strictly in the range of **0 to 255** ($2^8 = 256$

total values). The value **300 exceeds 255**, making it mathematically impossible for an 8-bit binary number to represent.

### 2. What a Valid IP Address Looks Like:

In a standard home/office local network using the 192.168.1.0/24 subnet (subnet mask 255.255.255.0):

- The first three octets (192.168.1) identify the network portion and must remain fixed.
- The last octet represents the host portion and must be a valid integer between **1 and 254** (since .0 is reserved for Network ID and .255 is reserved for Broadcast ID).
- **Valid Example IP Address:** 192.168.1.30 (or any unassigned IP between 192.168.1.1 and 192.168.1.254).